

Marcus R. Williams

DIRECTOR OF DATA SCIENCE & AI

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✦ Chicago, IL · Open to Remote

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Core Expertise

MACHINE LEARNING

Deep Learning / LLMs	●●●●●
NLP & Text Mining	●●●●●
Computer Vision	●●●●○
Reinforcement Learning	●●●●○

PROGRAMMING

Python	●●●●●
R / Julia	●●●●○
SQL / Spark	●●●●●
Scala	●●●○○

PLATFORMS

Databricks	●●●●●
AWS SageMaker	●●●●○
GCP Vertex AI	●●●●○
Azure ML	●●●○○

Education

Ph.D., Statistics

University of Chicago

2013 · Dissertation: "Scalable Bayesian Inference for High-Dimensional Data"

M.S., Applied Mathematics

MIT

2009 · GPA 4.0

B.S., Mathematics & CS

Howard University

2007 · Summa Cum Laude

Awards

- MIT Technology Review 35 Innovators Under 35, 2019
- Forbes 30 Under 30 – Enterprise Technology, 2018
- NeurIPS Best Paper Award, 2020
- McKinsey Digital Excellence Award, 2016

Publications

18 peer-reviewed papers · 2,400+ citations · h-index: 14

Featured in *Nature Machine Intelligence*, *ICML*, *NeurIPS*, *KDD*

13+

YEARS EXPERIENCE

\$420M

VALUE GENERATED

65

Strategic Data Science leader with 13+ years transforming complex datasets into revenue-driving intelligence. Built and led world-class ML teams at McKinsey, Uber, and Microsoft. Expert in designing end-to-end ML platforms from research to production at scale. Author of 18 papers and 3 patents in probabilistic ML and causal inference.

Professional Experience

Director of Data Science

Feb 2022 – Present

Microsoft Corporation · Redmond, WA (Remote)

- Lead 45-person data science organization spanning ML Engineering, Applied Research, and Analytics; \$38M operating budget
- Architected Microsoft Copilot's personalization engine serving 400M+ users across Office 365, increasing engagement by 34%
- Built company-wide ML feature store reducing model development lifecycle from 16 weeks to 3 weeks
- Drove \$180M incremental revenue through dynamic pricing models deployed across Azure Marketplace
- Established AI Ethics Review Board and led responsible AI framework adopted across all product divisions

Senior Principal Data Scientist

Aug 2018 – Jan 2022

Uber Technologies · San Francisco, CA

- Led surge pricing ML team; redesigned algorithm reducing driver idle time by 22% and increasing earnings by \$80M annually
- Built fraud detection system using Graph Neural Networks, preventing \$45M in fraudulent chargebacks in first year
- Created Uber's first causal inference platform enabling true lift measurement across 500+ growth experiments
- Launched ETA prediction model improving accuracy from 73% to 91% using deep sequence models

Engagement Manager - Advanced Analytics

Sep 2013 – Jul 2018

McKinsey & Company · Chicago, IL

- Led 12 data science engagements for Fortune 100 clients in retail, healthcare, and financial services; average ROI 8.4x
- Built predictive maintenance models for GE Aviation reducing unplanned downtime by 28% (\$65M savings)
- Designed customer lifetime value framework deployed at 3 major US banks, increasing retention by 19%

Technical Stack

PyTorch

TensorFlow

JAX

HuggingFace

LangChain

Apache Spark

dbt

Airflow

MLflow

Kubernetes

Docker

Snowflake

BigQuery

Tableau

Looker